

Reduced-Order Modelling

Scientific Machine Learning and Simulation Intelligence

Reduced-Order Modelling

Need of a computational **collaboration rather than **competition** between high and low order methods**

Simulation-based science is a constantly growing field for applied math and computational modelling.

Present and future efforts towards multi physics problems, as well as systems characterised by multiple spatial and temporal scales

Growing demand of:

- Efficient computational tools for
- Many query and real time computations
- Parametrised formulations
- Simulations of increasingly complex systems with uncertain scenarios

By industry and research partners

Reduced-Order Modelling

The answer: Reduced-order modelling

We will see that the mathematical and numerical formulation is so that reduced-order models can be applied to a wide range of scenarios characterised by PDEs of many different levels of complexity (from “simple” Poisson equations to strongly non-linear equations like Navier-Stokes equations).



Reduced-Order Modelling

The answer: Reduced-order modelling

In order to achieve real time computing, we need the so-called

Offline - Online

Computational paradigm the combines

**High Performance
Computing**



**Advanced Reduced Order
modelling techniques**



Reduced-Order Modelling

Reduced-order modelling in a nutshell

High/full order methods (FEM,FV,FD,SEM) to be accelerated using ROMs: $()^{\mathcal{N}}$

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μ (geometry, physical properties, etc.)

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Parametrised PDEs:

$$\mathcal{A}(u(\mu); \mu) = 0 \rightarrow \mathbf{A}^{\mathcal{N}}(\mu) \mathbf{u}^{\mathcal{N}}(\mu) = 0 \rightarrow \mathbf{A}_N(\mu) \mathbf{u}_N(\mu) = 0$$

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$$\mu \rightarrow s^{\mathcal{N}}(\mu) \rightarrow s_N(\mu)$$

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Reduced-Order Modelling

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High/full order methods (FEM,FV,FD,SEM) to be accelerated using ROMs: $(\cdot)^{\mathcal{N}}$

Reduced order model (ROM) which is the accelerator: $(\cdot)_N$

offline

very expensive pre-processing (high/full order): basic calculation (done **once**)

online

extremely fast (reduced order): real-time input-output evaluation

$$\mu \rightarrow s_N(\mu)$$

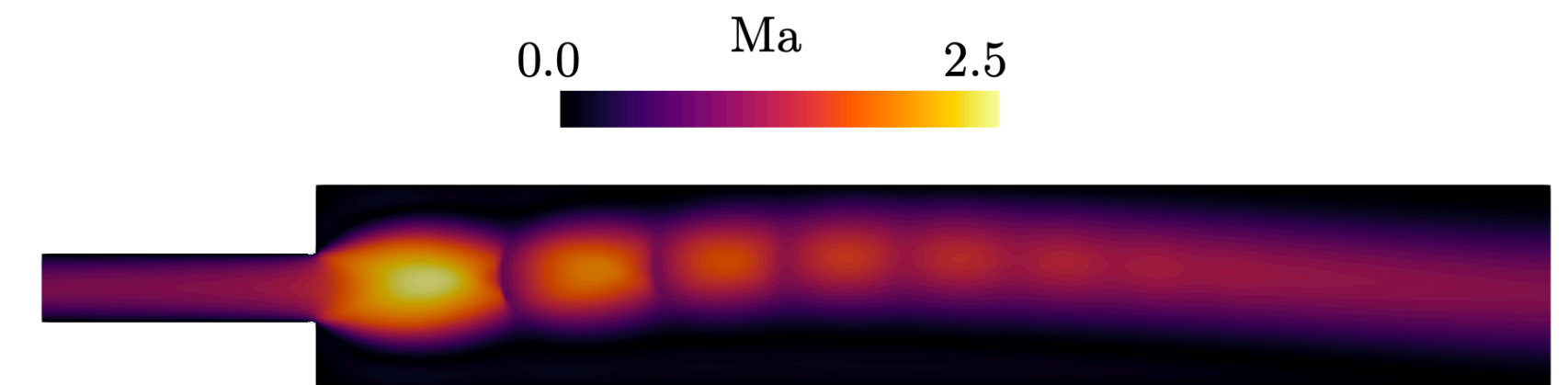
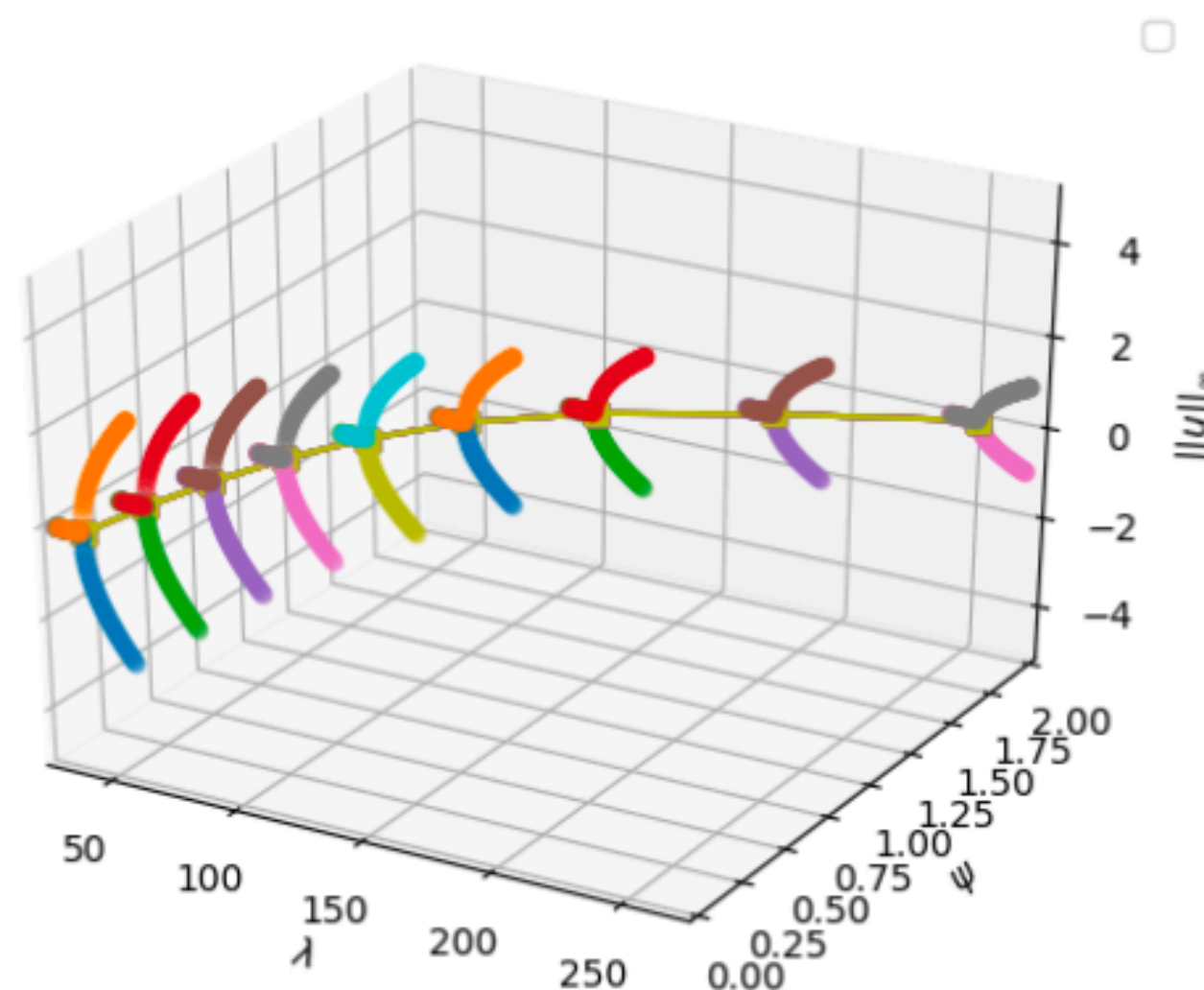
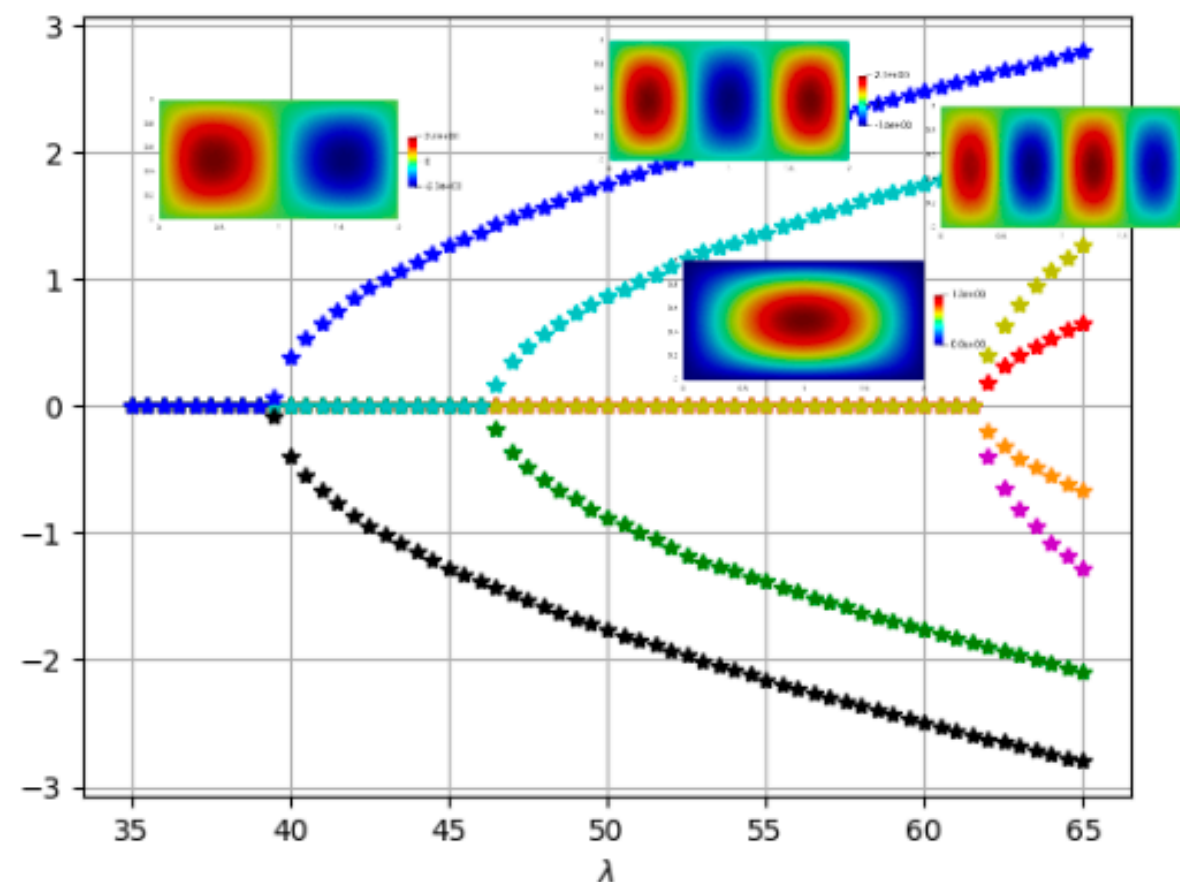
thanks to an efficient assembly of problem operators

challenges: stability, error bounds, efficient parametrisation, sampling

Reduced-Order Modelling

Bifurcation and stability analysis of parametrised problems by ROM

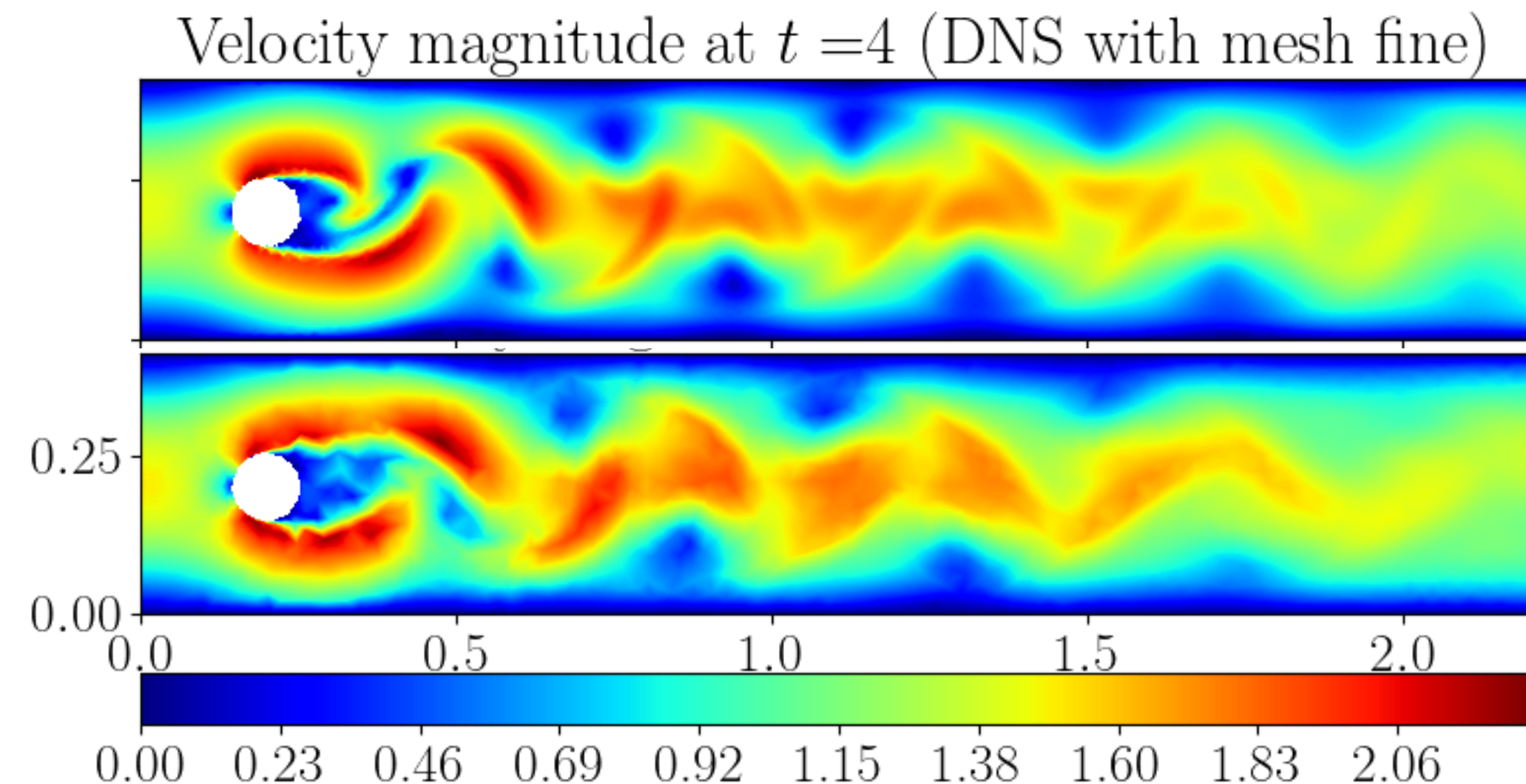
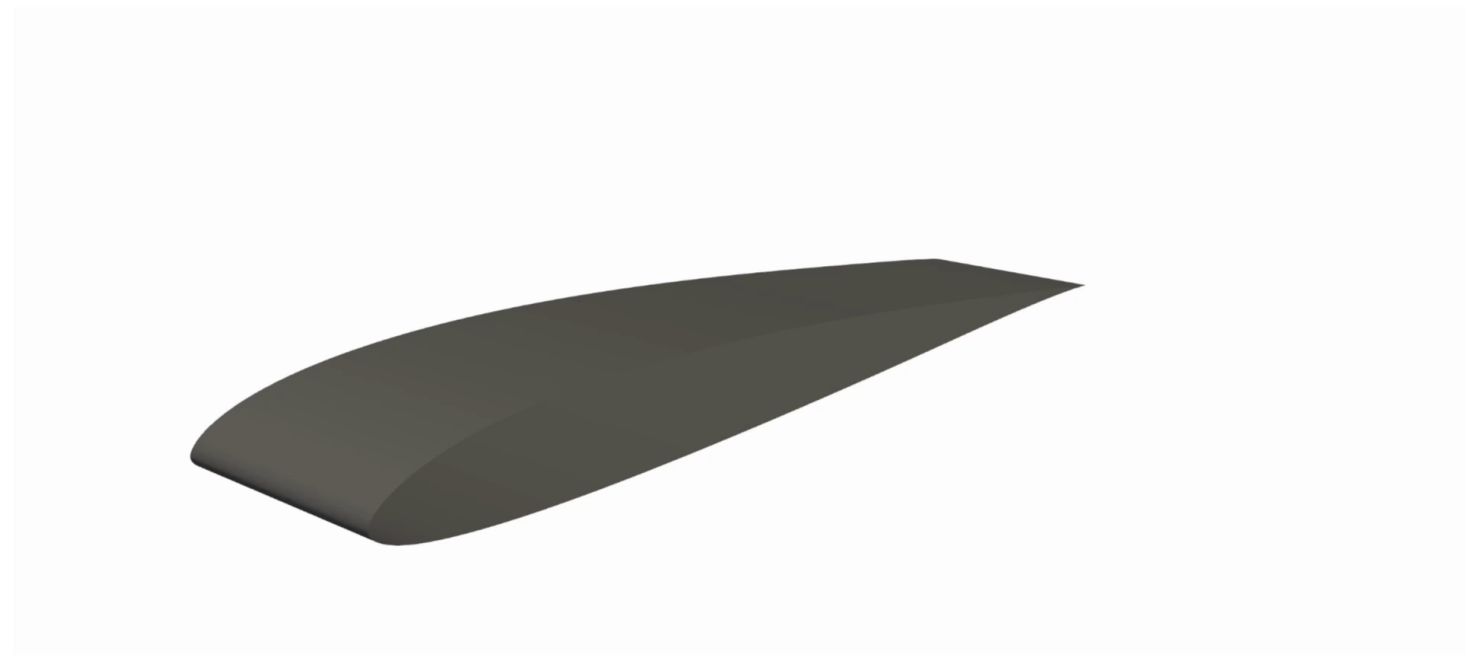
- *stability studies* in nonlinear problems are very expensive
- Understand and detect complex phenomena, such as *bifurcations*, leading to loss of uniqueness of the solutions also at the reduced order level
- Efficient reduced numerical techniques to detect *steady and Hopf bifurcations and branching* (continuation, eigenvalues analysis, efficient sampling, error bounds, a-posteriori error estimation)



Reduced-Order Modelling

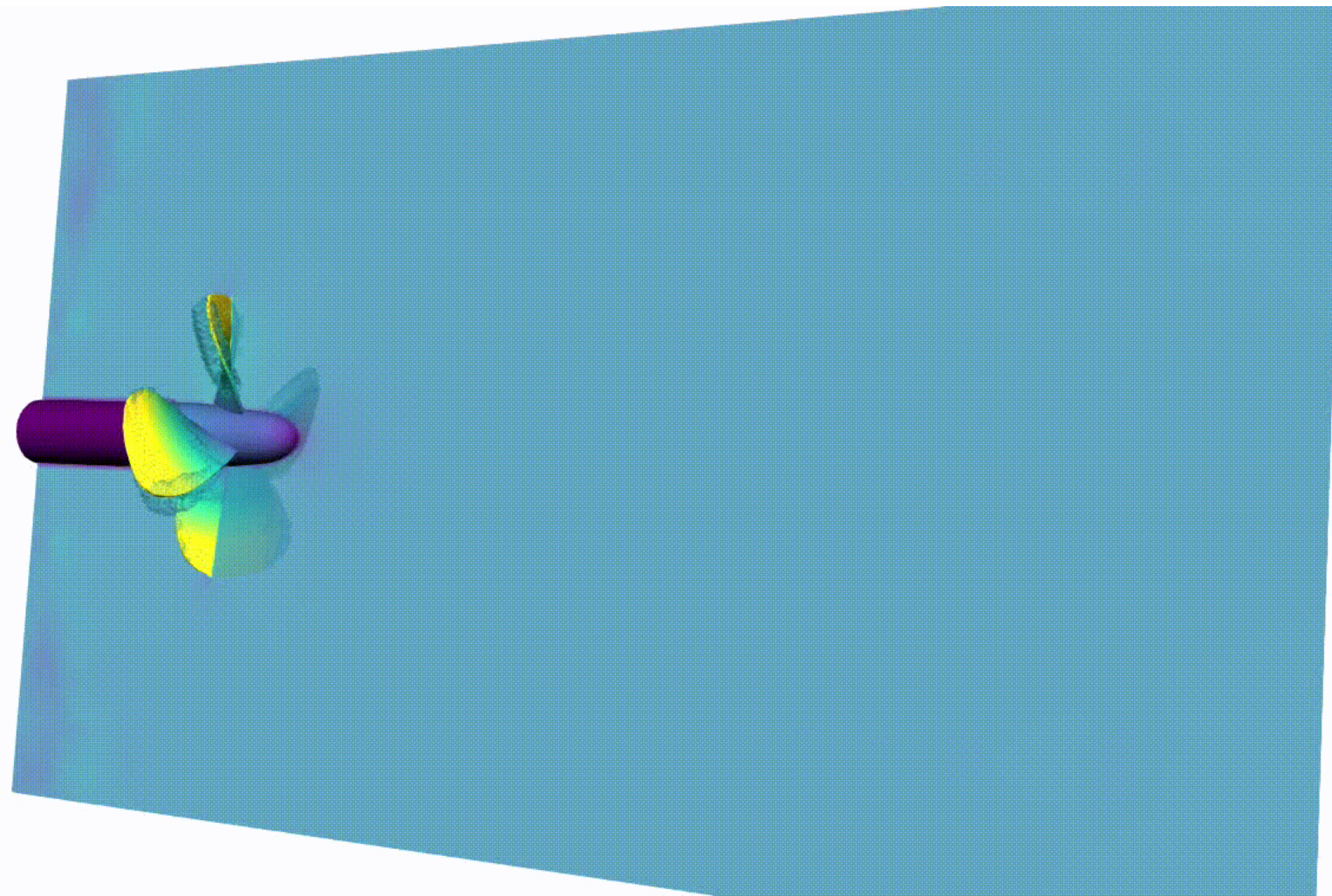
High Reynolds number flows with ROMs

- Many flows of engineering interest are characterised by advection dominated phenomena (e.g. turbulence)
- Reduced-order models are usually **unstable** for high-Reynolds numbers (turbulent flows)
- Offline-Online **stabilization techniques** for parametrised flows (geometry, physics) derived from streamline upwind Petrov-Galerkin (SUPG)
- A ROM **variational multiscale** approach in parametrised context to be properly developed towards turbulence modelling

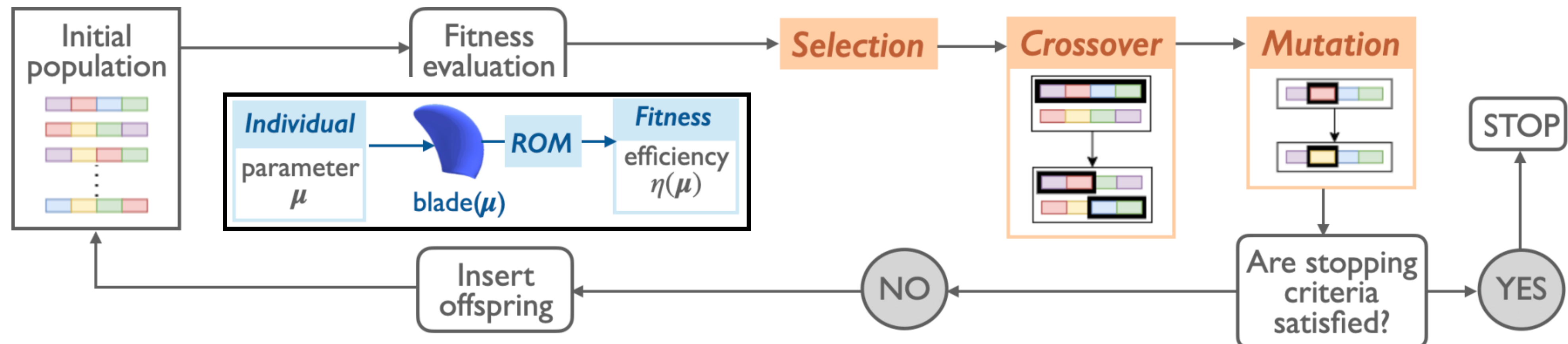


Reduced-Order Modelling

Shape optimization for marine propellers via ROMs



- Systems with complex geometrical parametrization
- Small efficiency improvements can lead to significant cost reduction
- Free-form deformation to explore the design space



Reduced-Order Modelling

Given $\mu \in \mathbb{P} \subset \mathbb{R}^P$

evaluate $s^e(\mu) = I(u^e(\mu))$,

where $u^e(\mu) \in X^e$ satisfies

$$a(u^e(\mu), v; \mu) = f(v), \quad \forall v \in X^e$$

The definitions used in the previous statement are:

μ : input parameter (P-tuple)

$\mathbb{P}$: input closed parameter domain in $\mathbb{R}^P$

s^e : output of interest form $\mathbb{P} \rightarrow \mathbb{R}$

I : linear bounded output functional

u^e : field variable from $\Omega \rightarrow \mathbb{R}^d$

X^e : functional space s.t. $(H_0^1(\Omega))^s \subset X^e \subset (H^1(\Omega))^d$

Reduced-Order Modelling

FEM

Let's then consider our heat equation in the simplest configuration (1D) and we want to discretise this using the Finite Element Method:

$$\cancel{\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}} \quad \kappa \frac{\partial^2 u}{\partial x^2} = f(x, \mu)$$

With FEM we start from the weak-formulation: we test the PDE against a test function and we then integrate:

$$\cancel{\int_0^1 v \frac{\partial u}{\partial t} dx = \kappa \int_0^1 v \frac{\partial^2 u}{\partial x^2} dx = 0.} \quad \kappa \int_0^1 v \frac{\partial^2 u}{\partial x^2} dx = \int_0^1 f(x, \mu) v dx.$$

Now, we consider the weak formulation of this equation which is

$$-\kappa \int_0^1 \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx = \int_0^1 f(x, \mu) v dx$$

Reduced-Order Modelling

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Reduced-Order Modelling

FEM

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$$-\kappa \int_0^1 \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx = \int_0^1 f(x, \mu) v dx$$

Which is the simple example of the general form:

$$a(u^e(\mu), v; \mu) = f(v), \quad \forall v \in X^e$$

Now, we are still living on the continuum level: we have not introduced any discretisation and (our solution is infinite dimensional). So, we look for a subset of our functional space: we look for solutions of the type

$$u(x) = \sum_{i=1}^{\mathcal{N}} u_i \phi_i(x)$$

For a general set of functions ϕ that we call basis functions. We then also set the test functions of the weak formulation to be like these ones and we get:

$$-\kappa \int_0^1 \frac{\partial \phi_j}{\partial x} \frac{\partial (\sum_{i=1}^{\mathcal{N}} u_i \phi_i)}{\partial x} dx = \int_0^1 f(x, \mu) \phi_j dx \quad \forall j = 1, \dots, \mathcal{N}$$

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Simplifying you get

$$-\kappa \sum_{i=1}^{\mathcal{N}} u_i \int_0^1 \frac{\partial \phi_j}{\partial x} \frac{\partial \phi_i}{\partial x} dx = \int_0^1 f(x, \mu) \phi_j dx \quad \forall j = 1, \dots, \mathcal{N}$$

In matrix form this is equal to

$$-\kappa \mathbf{K} \mathbf{u} = \mathbf{f}$$

Reduced-Order Modelling

In the classical FEM, we choose the approximation subspace as:

$$X^{\mathcal{N}} = \{v|_{T_h} \in \mathbb{P}_1(T_h), \forall T_h \in \mathcal{T}_h\} \cap X^e, \text{ such that } \dim(X^{\mathcal{N}}) = \mathcal{N}$$

Then we can formulate the FEM problem through the Galerkin projection:

Given $\mu \in \mathbb{P}$,

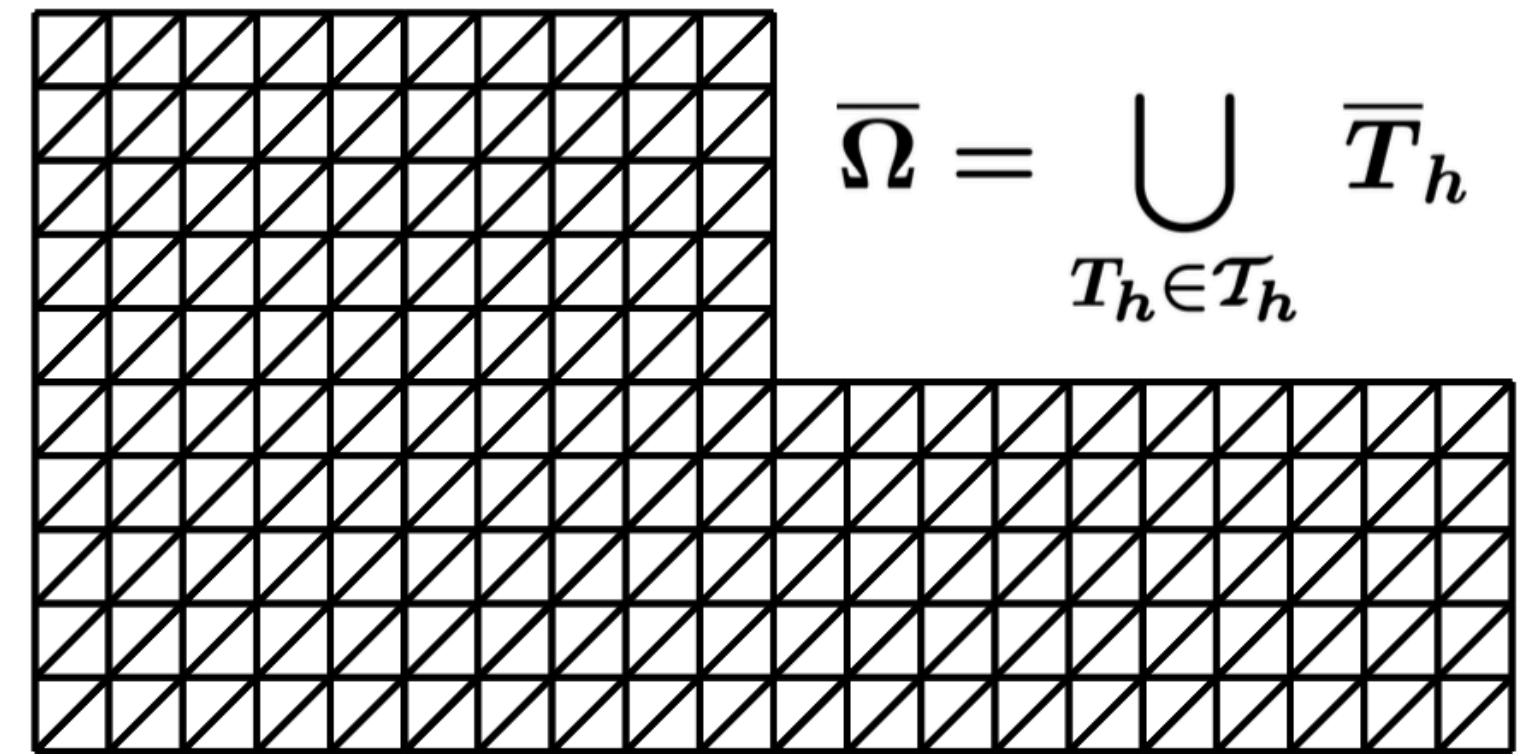
$$\text{evaluate } s^{\mathcal{N}}(\mu) = I(u^{\mathcal{N}}(\mu)),$$

where $u^{\mathcal{N}}(\mu) \in X^{\mathcal{N}}$ satisfies

$$a(u^{\mathcal{N}}, v; \mu) = f(v), \quad \forall v \in X^{\mathcal{N}}$$

Typically to obtain a good approximation of the output of interest a large value of $\mathcal{N}$ is needed so that

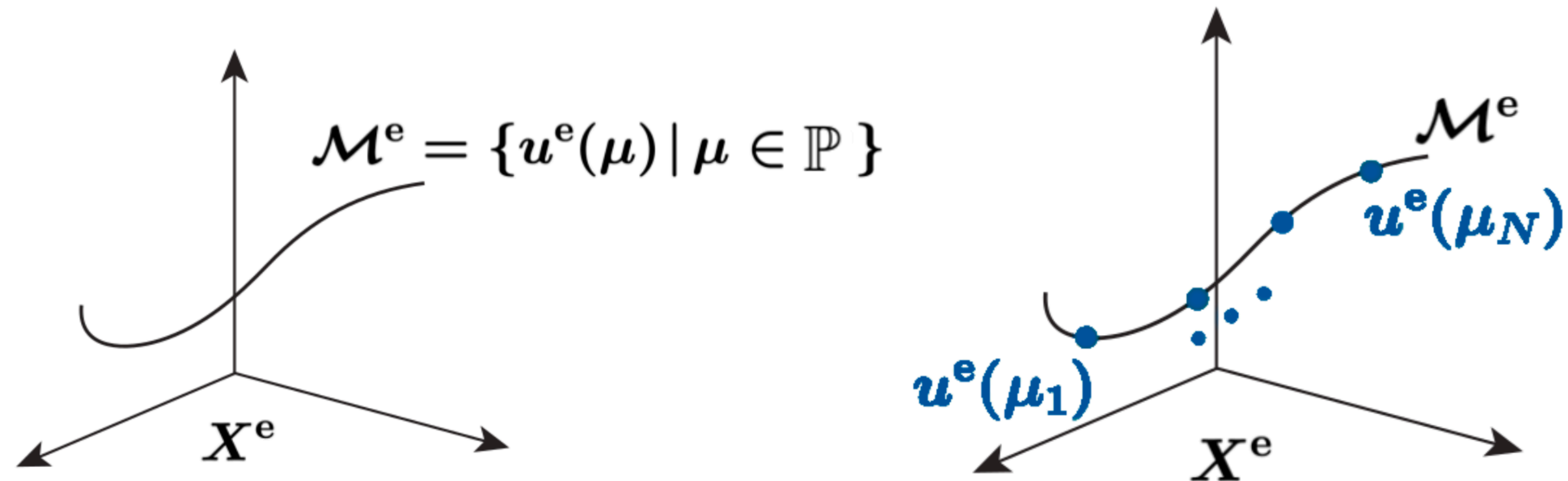
$$\mathcal{N} \gg 1 \rightarrow |s^e(\mu) - s^{\mathcal{N}}(\mu)| \ll 1$$



Reduced-Order Modelling

Once we have constructed a surrogate for $s^e(\mu), u^e(\mu)$, namely the FEM quantities $s^{\mathcal{N}}(\mu), u^{\mathcal{N}}(\mu)$

Let us introduce the notion of solution manifold comprising of all solutions of the parametric problem under variation of the parameter, i.e.



and the discrete version of the solution manifold $\mathcal{M}^{\mathcal{N}} = \{u^{\mathcal{N}}(\mu) \mid \mu \in \mathbb{P}\}$

Reduced-Order Modelling

We define an opportune sample $S_N = \{\mu_1, \dots, \mu_N \in \mathbb{P}\}$ and the hierarchical spaces $W_N = \text{span}\{\zeta_n = u^{\mathcal{N}}(\mu_n), 1 \leq n \leq N\}$, i.e., such that $W_N \subset W_{\bar{N}}$ if $N \leq \bar{N}$

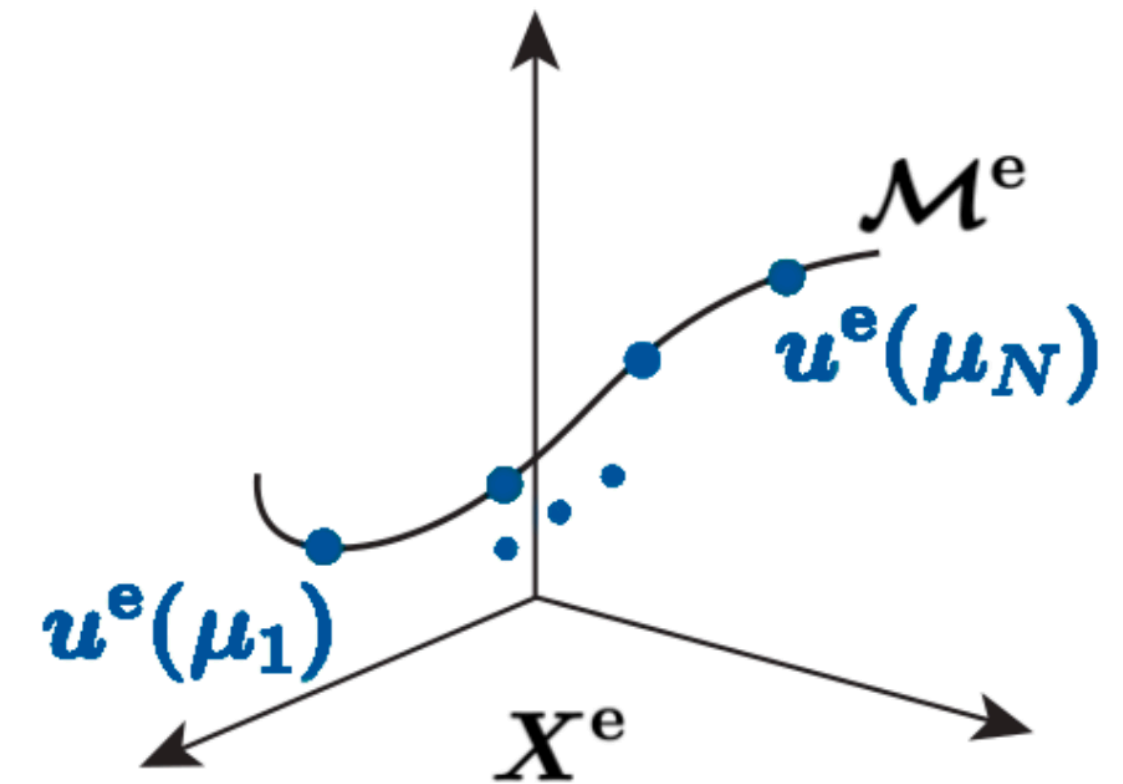
The reduced order model problem reads:

Given $\mu \in \mathbb{P}$,

$$\text{evaluate } s^N(\mu) = I(u^N(\mu))$$

where $u^N(\mu) \in W_N$ satisfies

$$a(u^N(\mu), v; \mu) = f(v), \quad \forall v \in W_N$$



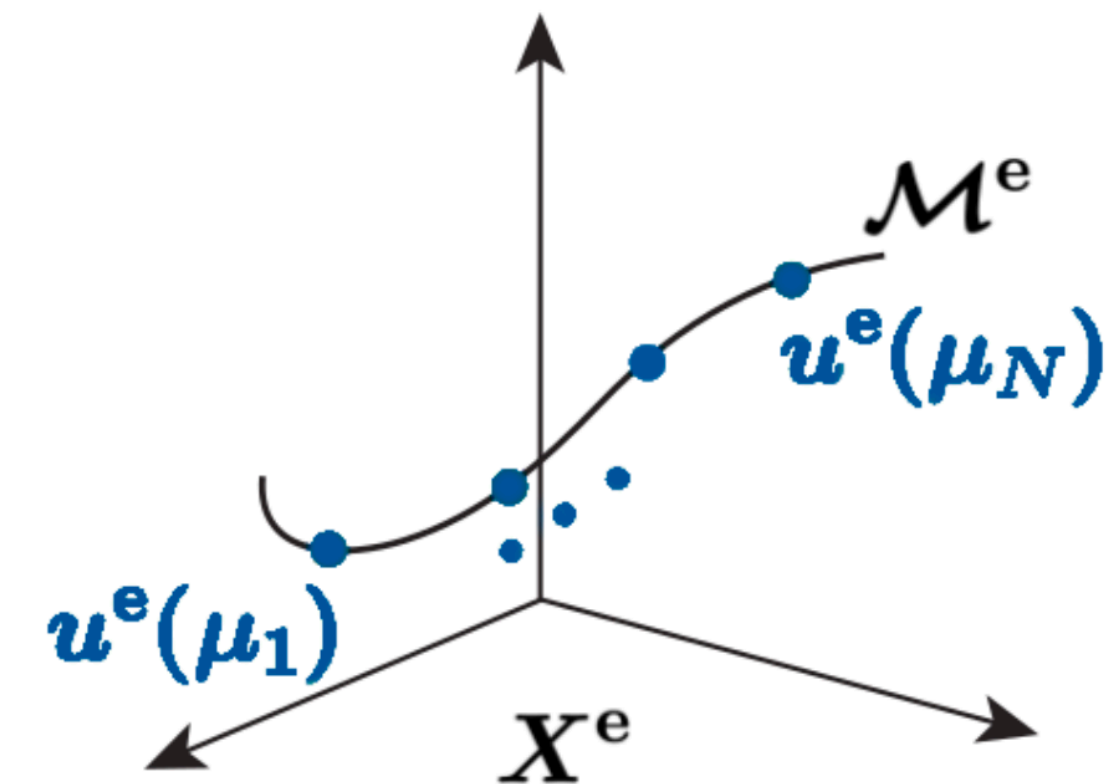
Reduced-Order Modelling

We can express every element of the space W_N in terms of its own base

$$u^N(\mu) = \sum_{j=1}^N u_j^N(\mu) \zeta_j$$

then we can evaluate $s^N(\mu) = \sum_{j=1}^N u_j^N(\mu) I(\zeta_j)$, where

$$\sum_{j=1}^N a(\zeta_j, \zeta_i; \mu) u_j^N = f(\zeta_i), \quad 1 \leq i \leq N$$



The offline/online procedure is the key point for the computational efficiency, in fact for the evaluation of $s^N(\mu)$, given u_j^N for $1 \leq j \leq N$:

- Offline: with computational cost $O(\mathcal{N})$
 - compute ζ_j , for $1 \leq j \leq N$;
 - Form and store $f(\zeta_i)$, for $1 \leq i \leq N$;
 - Form and store $a(\zeta_j, \zeta_i)$, for $1 \leq i, j \leq N$
- Online: with computational cost $O(N)$
 - perform sum $s^N(\mu) = \sum_{j=1}^N u_j^N I(\zeta_j)$

Reduced-Order Modelling

- Offline: with computational cost $O(\mathcal{N})$
 - Form and store $a(\zeta_j, \zeta_i)$, for $1 \leq i, j \leq N$

In the evaluation process for u_j^N , for $1 \leq j \leq n$, we can note that since

$$\zeta_j = \sum_{k=1}^{\mathcal{N}} (\zeta_j)_k \phi_k^{FE},$$

then

$$a(\zeta_j, \zeta_i) = a\left(\sum_{k=1}^{\mathcal{N}} (\zeta_j)_k \phi_k^{FE}, \sum_{l=1}^{\mathcal{N}} (\zeta_i)_l \phi_l^{FE}\right)$$

$$= \sum_{k=1}^{\mathcal{N}} \sum_{l=1}^{\mathcal{N}} (\zeta_j)_k a(\phi_k^{FE}, \phi_l^{FE}) (\zeta_i)_l$$

$$= \mathbf{Z}^T \mathbf{A}^{FE} \mathbf{Z}$$

The diagram illustrates the matrix equation $\mathbf{Z}^T \mathbf{A}^{FE} \mathbf{Z} = \mathbf{A}_q^N$. It consists of four colored boxes arranged horizontally: a light green box containing $\mathbf{Z}^T$, a light red box containing $\mathbf{A}^{FE}$, a light green box containing $\mathbf{Z}$, and a light green box containing $= \mathbf{A}_q^N$. The boxes are separated by spaces, and the entire expression is centered.

Reduced-Order Modelling

FEM

If we go back to our problem, where everything is basically the same but now we have a different basis

$$-\kappa \int_0^1 \frac{\partial \zeta_j}{\partial x} \frac{\partial (\sum_{i=1}^N u_i \zeta_i)}{\partial x} dx = \int_0^1 f(x, \mu) \zeta_j dx \quad \forall j = 1, \dots, N$$

Simplifying you get

$$-\kappa \sum_{i=1}^N u_i \int_0^1 \frac{\partial \zeta_j}{\partial x} \frac{\partial \zeta_i}{\partial x} dx = \int_0^1 f(x, \mu) \zeta_j dx \quad \forall j = 1, \dots, N$$

In matrix form you get

$$-\kappa \mathbf{K}^N \mathbf{u}^N = \mathbf{f}^N$$

Reduced-Order Modelling

To make a connection between Full-order and Reduced-order model we have basically the same thing we saw before

$$-\kappa \mathbf{K} \mathbf{u} = \mathbf{f}$$

The operator simply becomes:

$$-\kappa \mathbf{Z}^T \mathbf{K} \mathbf{Z} \mathbf{u}^N = \mathbf{Z}^T \mathbf{f}$$

And you end up with a smaller system of the type:

$$-\kappa \mathbf{K}^N \mathbf{u}^N = \mathbf{f}^N$$

Reduced-Order Modelling

The Proper Orthogonal Decomposition (POD) is an explore-and-compress strategy in which one samples the parameter space, compute the corresponding truth solutions at all sample points and following compression, retains only the essential information

Given Γ_{train} , we can define:

$$X_N^{\mathcal{N}, POD} = \arg \inf_{X_N^{\mathcal{N}} \subset \text{span}\{u^{\mathcal{N}}(\mu) | \mu \in \Gamma_{train}\}} \|u^{\mathcal{N}} - \Pi_{X_N^{\mathcal{N}}} u^{\mathcal{N}}\|_{L^2(\Gamma_{train}; X)},$$

Reduced-Order Modelling

We define $C^{POD} \in \mathbb{R}^{n_{train}} \times \mathbb{R}^{n_{train}}$ as:

$$C_{ij}^{POD} = \frac{1}{n_{train}} (u(\mu_{train}^i), u(\mu_{train}^j))_X, \quad 1 \leq i, j \leq n_{train}$$

Then we look at the eigenproblem

$$C^{POD} \zeta^k = \lambda^k \zeta^k$$

where we search for the eigenpairs $(\zeta^k \in \mathbb{R}^{n_{train}}, \lambda^k \in \mathbb{R}^{+,0}), 1 \leq k \leq n_{train}$.

Arranging the eigenvalues in discending order

$$\lambda^1 \geq \lambda^2 \geq \dots \geq \lambda^{n_{train}} \geq 0,$$

we can now identify a new basis in ζ_j

Moreover, we can define $N_{\max}$ as the smallest N such that

$$\epsilon_N^{POD} = \sqrt{\sum_{k=N+1}^{n_{train}} \lambda^k} \leq \epsilon_{\min}$$

The POD-RB spaces are $X_N^{POD} = \text{span}\{\zeta^n, 1 \leq n \leq N\}, 1 \leq N \leq N_{\max}$ with $(\zeta_n, \zeta_m)_X = \delta_{nm}, \quad 1 \leq n, m \leq n_{train}$

Reduced-Order Modelling

It can be proven that among all the possible basis of reduced dimensionality, the one provided by the POD is the optimal one that minimises the projection error

$$X_N^{\mathcal{N}, POD} = \arg \inf_{X_N^{\mathcal{N}} \subset \text{span}\{u^{\mathcal{N}}(\mu) | \mu \in \Gamma_{train}\}} \|u^{\mathcal{N}} - \Pi_{X_N^{\mathcal{N}}} u^{\mathcal{N}}\|_{L^2(\Gamma_{train}; X)},$$

The modes are ordered by importance:

$$\mathcal{C}^{\mathcal{N}} \zeta_i^{\mathcal{N}} = \lambda_i^{\mathcal{N}} \zeta_i^{\mathcal{N}}, \quad \lambda_1^{\mathcal{N}} \geq \lambda_2^{\mathcal{N}} \geq \dots \geq 0.$$

You then keep only the first N modes:

$$X_N^{\mathcal{N}, POD} = \text{span} \{ \zeta_1^{\mathcal{N}}, \dots, \zeta_N^{\mathcal{N}} \}.$$

You can then prove that the projection error is related to the eigenvalues of the neglected modes

$$\left\| u^{\mathcal{N}} - \Pi_{X_N^{\mathcal{N}, POD}} u^{\mathcal{N}} \right\|_{L^2(\Gamma_{train}; X)}^2 = \sum_{i=N+1}^{\mathcal{N}} \lambda_i^{\mathcal{N}}.$$

Reduced-Order Modelling

In previous sampling strategies we decided a-priori a dimension of the reduced basis and sampled in the parameter domain

BUT, we can do better than that and choose parameters to be sampled adaptively

Let's start from Γ_{train} , we can define $S_1 = \{\mu^1\}$, $W_1^{\mathcal{N}} = span\{u^{\mathcal{N}}(\mu^1)\}$

for $N = 2, \dots, N_{max}$:

$$\mu^N = \operatorname{argmax}_{\mu \in \Gamma_{train}} \|u^{\mathcal{N}}(\mu) - u_{N-1}^{\mathcal{N}}(\mu)\|_X$$

$$S_N = S_{N-1} \cup \mu^N$$

$$W_N^{\mathcal{N}} = W_{N-1}^{\mathcal{N}} \cup span\{u^{\mathcal{N}}(\mu^N)\}$$

end



we need n_{train} evaluations of the finite element solutions at each iteration!

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for $N = 2, \dots, N_{max}$:

$$\mu^N = \operatorname{argmax}_{\mu \in \Gamma_{train}} \Delta_{N-1}(\mu)$$

$$S_N = S_{N-1} \cup \mu^N$$

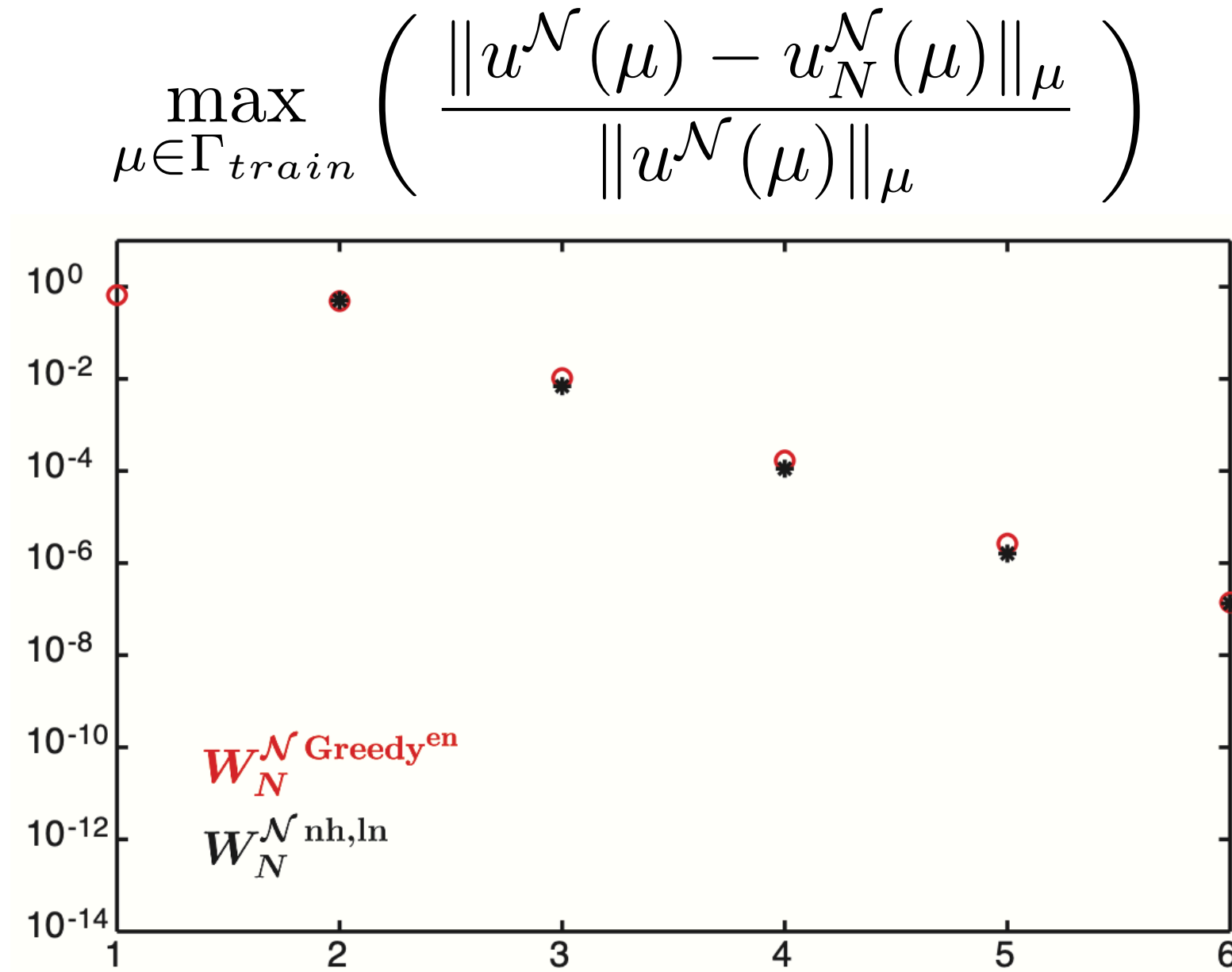
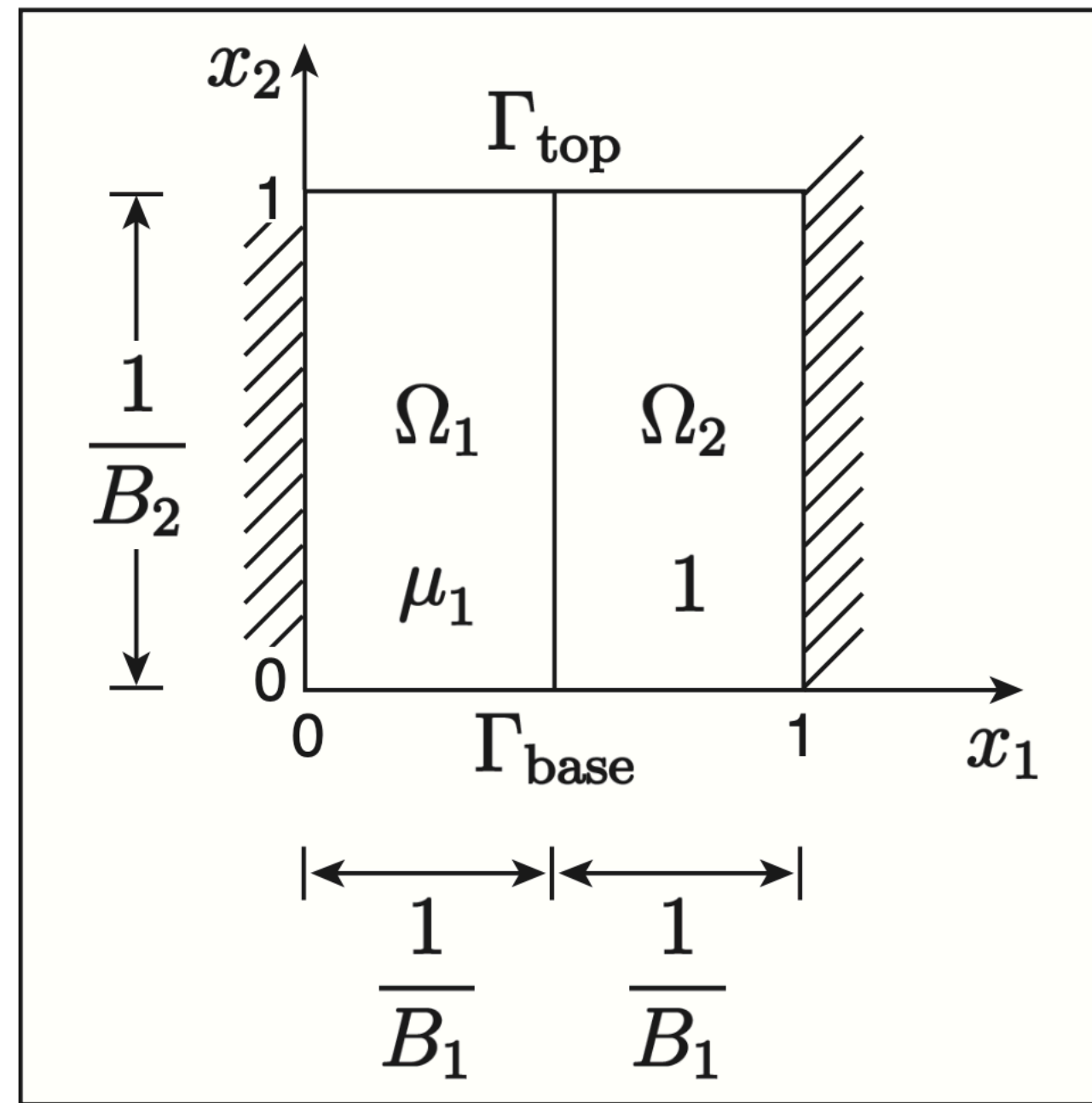
$$W_N^{\mathcal{N}} = W_{N-1}^{\mathcal{N}} \cup span\{u^{\mathcal{N}}(\mu^N)\}$$

$$\|u^{\mathcal{N}}(\mu) - u_N^{\mathcal{N}}(\mu)\|_X \leq \Delta_N(\mu), \quad \forall \mu \in \mathbb{P}$$

end

where $\Delta_N(\mu)$ is a sharp, inexpensive a-posteriori error bound for $\|u^{\mathcal{N}}(\mu) - u_N^{\mathcal{N}}(\mu)\|_X$

Reduced-Order Modelling



The Greedy algorithm heuristically minimizes RB error bound, while POD truly minimizes the projection error

However, the cost of Greedy is much lower than the cost of POD for large n_{train}

Reduced-Order Modelling

FEM

Let's try to make things a bit more complicated and we consider a time-dependent, non-linear case

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \kappa \frac{\partial^2 u}{\partial x^2} + f(x, \mu)$$

The weak form becomes

$$\int_0^1 \frac{\partial u}{\partial t} v \, dx + \int_0^1 u \frac{\partial u}{\partial x} v \, dx = \int_0^1 \kappa \frac{\partial^2 u}{\partial x^2} v \, dx + \int_0^1 f(x, \mu) v \, dx$$

And if we integrate by parts we get

$$\int_0^1 \frac{\partial u}{\partial t} v \, dx + \int_0^1 u \frac{\partial u}{\partial x} v \, dx = - \int_0^1 \kappa \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} \, dx + \int_0^1 f(x, \mu) v \, dx$$

Reduced-Order Modelling

FEM

Let's first consider the standard FEM discretization:

$$u(x) = \sum_{i=1}^{\mathcal{N}} u_i \phi_i(x)$$

We obtain

$$\int_0^1 \frac{\partial(\sum_{i=1}^{\mathcal{N}} u_i \phi_i)}{\partial t} \phi_j dx + \int_0^1 \left(\sum_{i=1}^{\mathcal{N}} u_i \phi_i\right) \frac{\partial(\sum_{i=1}^{\mathcal{N}} u_i \phi_i)}{\partial x} \phi_j dx = - \int_0^1 \kappa \frac{\partial(\sum_{i=1}^{\mathcal{N}} u_i \phi_i)}{\partial x} \frac{\partial \phi_j}{\partial x} dx + \int_0^1 f(x, \mu) \phi_j dx$$

And it becomes something like this

$$\sum_{i=1}^{\mathcal{N}} \frac{\partial u_i}{\partial t} \int_0^1 \phi_i \phi_j dx + \sum_{i=1}^{\mathcal{N}} u_i \int_0^1 \frac{\partial(\sum_{i=1}^{\mathcal{N}} u_i \phi_i)}{\partial x} \phi_j dx = - \sum_{i=1}^{\mathcal{N}} u_i \int_0^1 \kappa \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} dx + \int_0^1 f(x, \mu) \phi_j dx$$

Reduced-Order Modelling

FEM

$$\sum_{i=1}^{\mathcal{N}} \frac{\partial u_i}{\partial t} \int_0^1 \phi_i \phi_j dx + \sum_{i=1}^{\mathcal{N}} u_i \int_0^1 \frac{\partial(\sum_{i=1}^{\mathcal{N}} u_i \phi_i)}{\partial x} \phi_j dx = - \sum_{i=1}^{\mathcal{N}} u_i \int_0^1 \kappa \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} dx + \int_0^1 f(x, \mu) \phi_j dx$$

In matrix form you get

$$\mathbf{M} \frac{d\mathbf{u}}{dt} + \mathbf{C}(\mathbf{u})\mathbf{u} = \mathbf{K}\mathbf{u} + \mathbf{f}$$

First, we have to deal with the non-linear term in standard FEM. How we can do that? We have different options but the simplest is using explicit Euler

$$\mathbf{M} \frac{\mathbf{u}^{t+1} - \mathbf{u}^t}{\Delta t} + \mathbf{C}(\mathbf{u}^t)\mathbf{u}^t = \mathbf{K}\mathbf{u}^t + \mathbf{f}$$

The linear system to solve is easier and we get away with it

Reduced-Order Modelling

FEM

$$\mathbf{M} \frac{\mathbf{u}^{t+1} - \mathbf{u}^t}{\Delta t} + \mathbf{C}(\mathbf{u}^t) \mathbf{u}^t = \mathbf{K} \mathbf{u}^t + \mathbf{f}$$

From this one we can simply use a change of basis and get the same for the ROM

$$\mathbf{Z}^T \mathbf{M} \mathbf{Z} \frac{\mathbf{u}_N^{t+1} - \mathbf{u}_N^t}{\Delta t} + \mathbf{Z}^T \mathbf{C}(\mathbf{Z} \mathbf{u}_N^t) \mathbf{Z} \mathbf{u}_N^t = \mathbf{Z}^T \mathbf{K} \mathbf{Z} \mathbf{u}_N^t + \mathbf{Z}^T \mathbf{f}$$

The non-linearity breaks the ROM standard approach and forces you to operate on the high-dimensional space. This implies the need for what is commonly referred to as Hyper-reduction or to avoid projection based ROMs and go straight to non-intrusive ROMs